

Naveen khan

AI Engineer

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PROFILE

AI Engineer with hands-on experience building production-ready AI systems, including RAG-based applications, LLM-powered chatbots, and computer vision solutions. Skilled in developing scalable backend architectures using FastAPI and integrating AI systems into real-world platforms, including web applications. Experienced in developing semantic retrieval and Text-to-SQL solutions for real-world applications. Passionate about applying AI to solve practical problems across enterprise and industrial domains.

WORK EXPERIENCE

Full Stack AI Engineer – AI Hackathon, Sofstica Solutions (Pvt.) Ltd 08/2026

- Built a clinical research platform with **Text-to-SQL** capabilities, and engineered an end-to-end data Quality Engine from scratch across **912,284 patient records** with full provenance tracking.
- Integrated a FastAPI + Next.js app featuring **LLM-powered natural language interfaces** supporting real-time cohort exploration, analysis, and AI-generated summaries for faster data-driven research.

Tools: Python, FastAPI, Next.js, SQLite, LLM, LangChain, Docker, Azure

AI Engineer Intern – Itsolera Pvt. Ltd. 01/2026 – 04/2026

- Trained CNN and YOLO-based Computer Vision models on **10,000+ images spanning three domains**: medical image classification, damaged road detection, and theft detection.
- Fine-tuned models achieving **93%+ accuracy** through systematic hyperparameter optimization and various other techniques. Automated repetitive data workflows, cutting manual processing time by approximately **40%** across multiple concurrent projects.

Tools: Python, Deep learning Frameworks, FastAPI, Streamlit, Roboflow, Computer Vision

AI Engineer Intern – Civil Aviation Authority of Pakistan (CAA) 07/2025 – 08/2025

- Engineered an enterprise-level conversational system using **Large Language Models (LLMs)** that reduced internal query resolution time by **60%**.
- Architected a semantic retrieval framework leveraging **Retrieval-Augmented Generation (RAG)** and embedding-based document indexing across **100+ organizational PDFs**.
- Collaborated with technical teams on AI workflows, data preprocessing, and model integration for enterprise-level applications.

Tools: Python, FastAPI, LangChain, RAG, NLP, LLMs, Hugging Face, .NET

AI Research - Multimodal Smart Wearable for Personal Safety 11/2024 – 11/2025

- Developed an AI-powered wearable with **YOLO-based robbery detection**, multilingual speech recognition, evidence capturing, and GPS/GSM emergency response.
- Achieved **95% accuracy** in real-time threat detection with alert response time under **5 seconds**.
- Achieved **2nd Place** at IEEE CS Exhibition 2025 (**among 45+ projects**), and published a **research paper globally in 2026**.

Tools: Python, PyTorch, YOLOv8, Raspberry Pi 4, ESP32, IoT, GPS/GSM, Computer Vision

SKILLS

- Python | Machine Learning | Seaborn | Computer Vision | LLMs | RAG | AI Agents | LangChain | Prompt Engineering
- Pandas | TensorFlow | FastAPI | Streamlit | Git | GitHub | LLM API Integration | Vector Databases (Pinecone, FAISS)
- MySQL | PostgreSQL | HTML | CSS | JS | n8n Automation | JSON | Workflow Automation | Docker | Vercel

EDUCATION

BE Computer Systems Engineering 11/2021 – 12/2025
Mehran University of Engineering & Technology, Pakistan

PUBLICATIONS

- Khan, N., et al. (2026). SAFELINK: A Multimodal Smart Wearable Device for Personal Safety. *THESES: International Journal of Multidisciplinary Research*. [Link]